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RAGUL SEETHARAMAN

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Professional Summary

AI/ML Engineer with 3 years of experience building agentic AI applications and production ML systems across healthcare, automotive, and finance, and dual Master's degrees from Virginia Tech. Recognized for contributions to an FDA-approved product, leadership as an NSF I-Corps Entrepreneurial Lead, winning Capital One's Best Financial Hack, and holding two patents alongside research publications

Education

Virginia Tech, USA *[**Dual Masters**] MSc in Computer Engineering & MSc in Biological Systems Engineering* **Aug 2023 – May 2026**
Vellore Institute of Technology, India, *BTech in Electronics and Communication Engineering* **Jul 2016 - May 2020**

Skills

Programming	Python, SQL, C++, MATLAB
Tools/Frameworks	TensorFlow, PyTorch, OpenCV, Pandas, RAG, LangChain, LangGraph, FastAPI
Data/Cloud	Databricks, AWS (SageMaker, EC2, S3), Airflow, Docker, Kubernetes, Terraform (Infrastructure as Code), PostgreSQL, Parquet, PyArrow, CI/CD, MLOps, pipeline orchestration, logging and monitoring
Coursework	Advanced Machine Learning, Applications of Machine Learning, Data Engineering, Database Systems & Management, Natural Language Processing, Computer Vision

Experience

Cotiviti, Utah (USA) | **Healthcare Payment Accuracy** **Jun 2025 – Present**
Generative AI Research Engineer Intern

- Developed a three-tier hybrid **RAG** pipeline for contract intelligence across **2,630 agreements**, using **PaddleOCR-VL** for page-level text extraction, semantic chunk segmentation, and **OpenAI text-embedding-3-large** for embedding generation, with projected savings of up to **$4M annually**.
- Built hybrid semantic retrieval using **Databricks Vector Search**, **BM25**, and adaptive rank fusion with scheduled indexing orchestrated **LangGraph** agentic workflows with stateful routing, retrieval retries, and **LLM-as-judge confidence gates** for tier selection.
- Implemented entity resolution, multi-hop contract QA, and citation tracing across multiple contract types; evaluated using **Arize Phoenix**, achieving **81% QA correctness**, **84.6% groundedness**, and **0.89 NDCG@10** monitoring to investigate retrieval failures.
- Exposed the RAG pipeline through **FastAPI RESTful APIs**, implementing centralized authorization, TLS validation, and **Guardrails AI** for prompt-injection detection and PII/PHI screening.
- Managed **Terraform**-based AWS infrastructure (**EC2, Load Balancer, Route 53**), containerized Python services with **Docker**, and supported **Kubernetes** deployments; maintained **CI/CD** pipelines for automated builds and releases, troubleshooting deployment and runtime failures through application and infrastructure logs.

Turmerik, New York (USA) | **Biotechnology** **Sep 2024 – Dec 2024**
Machine Learning Engineer Intern | *NSF I-CORPS*

- Developed a clinical trial protocol generation system using **LangChain**, indexing embeddings in **ChromaDB**, retrieving similar studies with **FAISS**, and using **ROUGE**-scoring evaluation to improve output quality from ClinicalTrials.gov data.
- Led customer discovery as Entrepreneurial Lead in the **NSF I-CORPS** program, engaging **25+ industry leaders** to align product development with clinical research needs.

AmberTAG Analytics (nference), USA (remote) | **Healthcare** **May 2021 - Nov 2022**
Data Analyst

- Improved arrhythmia detection to **94.2% accuracy** using **1D CNN** and **Xception** on ECG/EEG data, with **Fourier Transform** and **EMD** for noise reduction and contributed to **FDA-approval** of the product later acquired by nference.
- Supported a real-time cardiac-monitoring pipeline using **RabbitMQ** for asynchronous messaging and **PostgreSQL** for data storage, integrating **3D visualizations** tested on 11 patients, packaged ML services with **Docker** and maintained automated build workflows.

Cognizant, India | **E-commerce** **Sep 2020 - May 2021**
Programmer Analyst Trainee

- Built and deployed a recommendation pipeline for **6,000+ Shopify products** using **TF-IDF**, **Word2Vec**, and **K-means clustering** on **AWS SageMaker**, improving recommendation relevance by **15%** and conversion rate by **8%**.

Maruti Suzuki R&D, India | **Automotive** **Feb 2020 - Aug 2020**
Data Analyst Intern

- Applied **ANOVA** feature analysis to select **14 battery-performance parameters** and **PCA** to reduce a separate block of correlated battery-health signals for sequence modeling.
- Built **XGBoost** and **LSTM** training pipelines using **10,000+ vehicle sensor and battery records** from **fleet cars**. Forecast cranking voltage from per-vehicle histories and estimated remaining battery life from time to a voltage threshold.

Patents & Research

A NOVEL APPROACH USING DEEP AUTOENCODERS AND LARGE LANGUAGE MODELS FOR INTRUSION DETECTION **Aug 2024**
AAAI Fall Symposium 2024

- Trained a **PyTorch deep autoencoder** for network intrusion detection on **NSL-KDD**, compressing **108 traffic features to a 4-dimensional bottleneck** and achieving **90.6% accuracy**, **86.8% precision**, **98.4% recall**, and **92.3% F1** on anomaly detection.
- Built a two-stage intrusion detection pipeline using autoencoder reconstruction error for anomaly detection and **LLaMA 3.1-8B** for alert triage, severity ranking, and response recommendations.

ADVANCED SHOPPING TROLLEY WITH PRODUCT AUTHENTICATION AND FAST BILLING **Sep 2020**
Patent applications: 202041036031, 20204102288 | IEEE CONECCT | Larsen & Toubro National Finalist (Top 1.7%)

- Deployed **YOLOv2 Tiny and OpenCV** inference on **Raspberry Pi**, combining camera, barcode, and load-cell inputs for automated billing; reported **95.3% object-detection accuracy**.
- Built **Firebase** monitoring and logging for 40+ edge devices to help triage inference and sensor issues; added collaborative-filtering recommendations to a checkout system that reduced queue time.

Projects

Time-Aware LLM Sepsis Forecaster **Dec 2025**
AMIA 2025 Submission

- Built a multi-horizon clinical prediction model fine-tuning **Meditron-7B** and **BioMistral** with **4-bit QLoRA** (NF4, LoRA on attention + MLP projections) to predict sepsis onset at 6/12/24/48h from free-text ICU notes across **46K MIMIC-III** stays.
- Engineered a **PyTorch** training pipeline with **weighted BCE**, **focal**, and **asymmetric loss** plus positive oversampling to handle severe class imbalance, with per-horizon **AUROC/AUPRC/F1** evaluation and threshold tuning.
- Benchmarked fine-tuned 7B LLMs against a **BioMed-RoBERTa** baseline that reached **0.82 AUROC** and **0.46 recall**, showing a smaller supervised model can outperform larger medical LLMs on operating-point metrics under extreme imbalance.

Crediwise - Use the right card, every time. **Sep 2025**
MLH Hackathon - Capital One: Best Financial Hack | VT StartupSprint: Best Problem-Solution Fit

- Engineered a full-stack credit card recommendation platform using a **React** frontend, **Flask** backend, and **Manifest V3** Chrome extension, integrated with **Supabase** to deliver real-time checkout **card recommendations** and an interactive savings dashboard.
- Designed a LangGraph-based **multi-agent architecture** orchestrating specialized agents for transaction classification, reward optimization, and offer validation, integrated **Bright Data MCP** for tool-based retrieval and recommendations in live card offers.

UAV-Based Deep Learning for Multi-Crop Plant Counting **Jan 2025**
MS Thesis, Biological Systems Engineering, Virginia Tech

- Benchmarked **YOLOv12, YOLO26, RT-DETR, and RF-DETR** across four crop phenotypes using 2,481 image tiles and 95,989 annotations, achieving 0.90–0.96 mAP@0.5 with YOLOv12.
- Built a **tile-based inference pipeline** for high-resolution UAV GeoTIFF imagery, handling coordinate remapping, cross-tile NMS, and CRS transformations to generate plant counts, density heatmaps, and GIS outputs.
- Developed a **FastAPI** backend and **Streamlit** interface for uploading orthomosaics, running model inference, and exporting shapefiles, containerized both services using **Docker Compose** and deployed inference workflows on **AWS**.